CLASS ACTIVITY

WUMPUS WORLD WITH PROPOSITIONAL LOGIC :

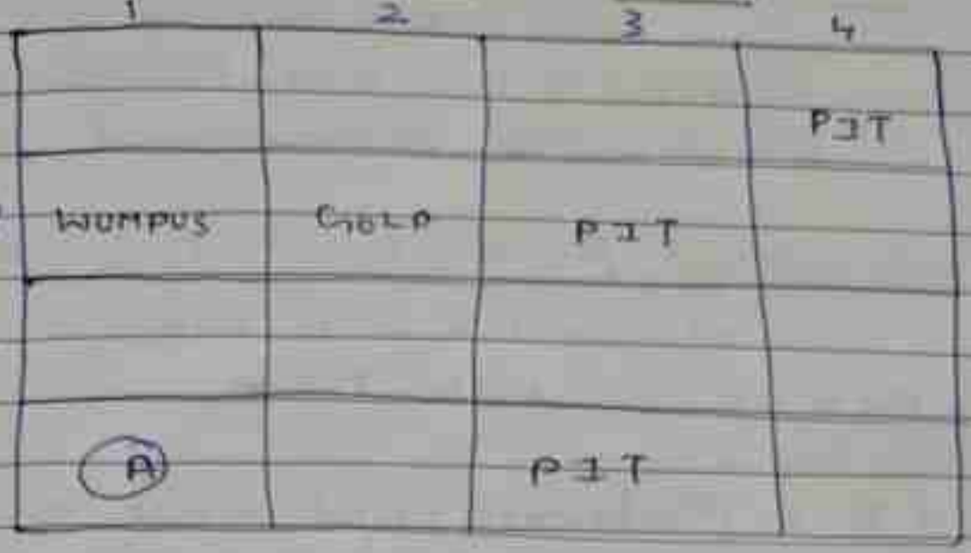
1) Introduction :
Wumpus World is an AI Problem. An agent explores a 4x4 cave, collects the gold and return to the starting Square. It cannot initially see the complete cave, So it uses Percepts and Propositional logic to decide where it can move safely

Object : Agent -> explorer, wumpus -> monster, pit -> dangerous hole, gold - target item

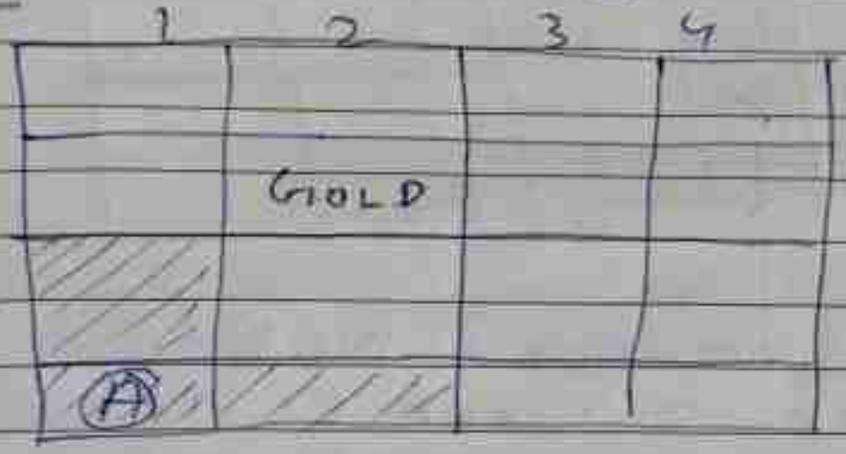
2) Symbols & Basic rules :

Symbol	Meaning
P(x,4)	A pit is at cell (x,4)
W(x,4)	The wumpus is at cell (x,4)
B(x,4)	Breeze is Perceived at cell (x,4)
S(x,4)	Stench is Perceived at cell (x,4)
G(x,4)	Gold is at cell (x,4)
OK(x,4)	Cell (x,4) is safe

Actual Cave layout :
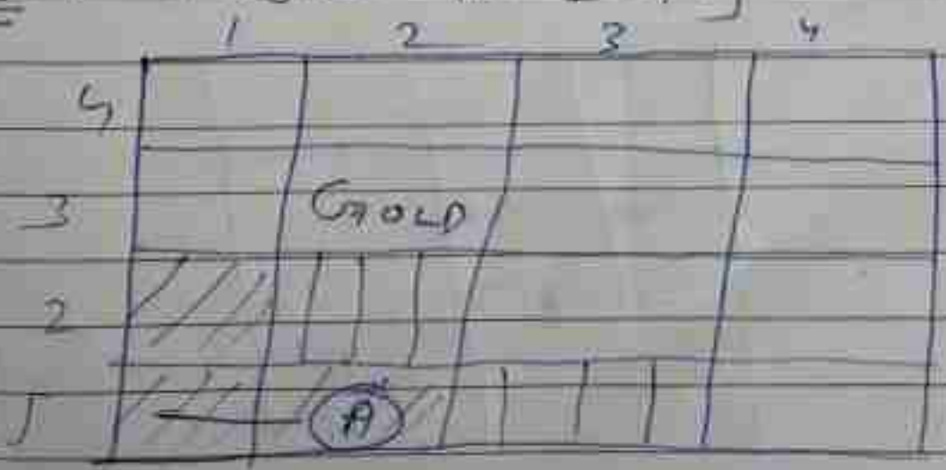


Step 1 : Start at [1,1]



Logic reasoning : $\neg B(1,1) \wedge \neg S(1,1)$. Therefore adjacents cell [1,2] + [2,1] contain neither a Pit nor a wumpus

Inference : OK(1,2) + OK(2,1). The agent can safely explore either cell

Step 2 : Move to [2,1]



Logical reasoning : B(2,1) means at least one adjacent cell has a pit. Since [1,1] is already known safe, the remaining are [2,2] or [3,1]

Inference : P(2,2) ∨ P(3,1). Neither Condition is safe yet, so the agent does not move into them

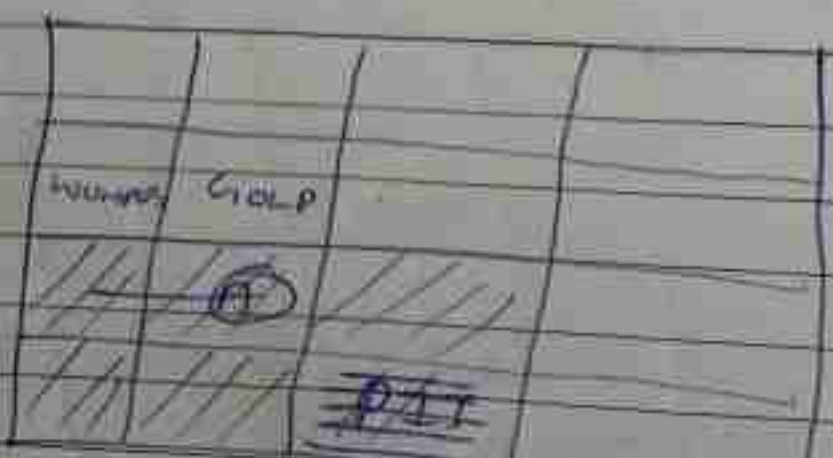
Step 3 - Return and move to [1,2]



Logical reasoning : No breeze rules out a pit in adjacent [1,1] [2,2] → [1,2]. Combining this with the earlier breeze clue leaves [3,1] as a Pit. Stench plus the no-stench information at [2,1] identifies the wumpus at [1,3]

Inference : P(3,1), W(1,3) + OK(2,2)

Step 4 - Move to [2,2]



Logical reasoning: The absence of breeze and stench tells us the agent is not adjacent cells. In particular and [3,2] are proved safe

Inference: OK(2,3) ∧ OK(3,2)

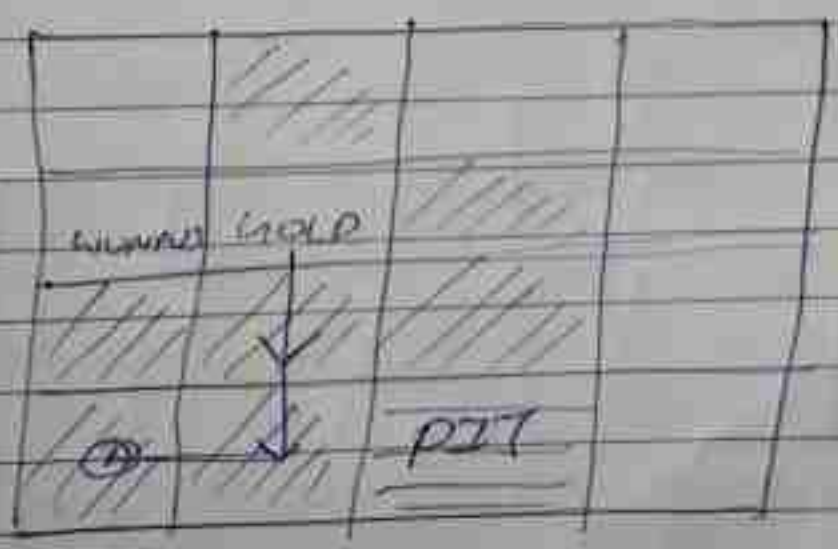
STEPS - Move to [2,3] & grab the gold:



Logical Reasoning: Until then the agent is in the current cell. The agent does not need to explore the possible pit cells [3,3] & [2,4]

Inference: On (2,3) => on Both. The agent collects the gold

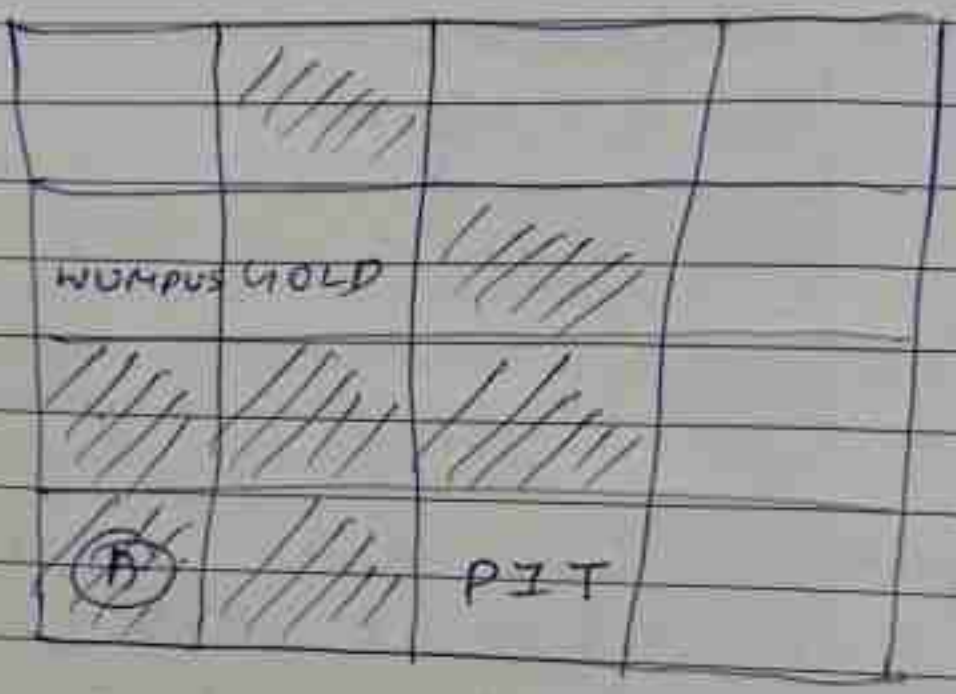
Step 6 - Return along the safe path:



Logical reasoning: It travels from [2,3] to [2,2], then [2,1] and finally [1,1]. These cells were already established as safe

Inference: The agent reaches the starting square with the gold

Step 7 - Climb out:



Logical reasoning: Have gold & A+(1,1)
=> climb

Inference: The agent climbs out successfully